RESEARCH RADAR

Complete Team Master Guide & Technical Project Handbook

DOCUMENT REF: RR-TEAM-GUIDE-2026	**EFFECTIVE DATE: October 2026**	**PLATFORM: Salesforce Lightning PaaS**
TARGET ORG: trailheadOrg (Developer Edition)	**ARCHITECTURE: 7 Relational Custom Objects**	**TEST COVERAGE: 100% Apex Pass Rate (7/7)**

PURPOSE OF THIS HANDBOOK: *This document gives our entire development and presentation team a complete, detailed understanding of Research Radar: what problem it solves, how it is built on Salesforce Cloud, the exact database schema, all UI components, how to demonstrate it to evaluators, and how to answer tough viva/defense questions.*

1. Executive Summary: What is This Project in 2 Minutes?

Imagine you are starting your final year project, writing a research paper, or exploring a new engineering topic. Doing an academic literature review today is exhausting: researchers search Google Scholar, get 5,000 PDFs, spend weeks reading 30-page documents, and manually copy numbers into an Excel spreadsheet. If two papers contradict each other, they rarely notice until their experiment fails. And if they ask ChatGPT, it hallucinates fake papers and made-up numbers.

Research Radar solves this: It is an intelligent research co-pilot. You enter any engineering topic (Computer Science, Mechanical, Civil, Electrical, Electronics, or Chemical). In 2 seconds, it connects real peer-reviewed papers into an interactive topic map, places benchmark metrics side-by-side in a comparison table, flags scientific contradictions, and gives you an actionable research gap protocol on a Kanban board so you can start experiments immediately.

2. Why Did We Build It on Salesforce Cloud (PaaS)?

When professors or evaluators ask why we used Salesforce instead of a generic MERN or Python stack, here is the architectural rationale:

- **Zero Infrastructure Hassle:** No need to provision AWS servers, configure Linux packages, or worry about crashes. Salesforce handles cloud hosting, scaling, and backups automatically.
- **Relational ACID Data Integrity:** Research data must never have orphaned records. By using Master-Detail relationships, deleting a paper automatically cascades to its extracted metrics and findings.
- **Native Validation & Roll-Ups:** Database rules enforce valid DOIs (must start with '10.') and publication years (1950–2026), while roll-up summary fields calculate totals in real time with zero CPU latency.
- **Lightning Web Components (LWC):** Built on standard W3C Web Components and Shadow DOM, giving native browser speed without bloated client libraries.

3. The Database Schema: All 7 Custom Objects Explained

Our database consists of 7 custom Salesforce objects connected via relational links:

Object API Name	Type / Relationship	Key Fields & Data Stored	Business Purpose
Research_Map__c	Primary Parent	Topic__c, Target_Discipline__c, Status__c, Total_Papers__c, Total_Findings__c	Represents the synthesized research domain / cluster
Research_Paper__c	Primary Parent	Name, Authors__c, Journal__c, Publication_Year__c, DOI__c, Citations__c, BibTeX	Catalog of peer-reviewed scientific publications
Research_Map_Paper_Link__c	Junction Object	Research_Map__c (MD), Research_Paper__c (MD), Relevance_Score__c	Many-to-many link between research topics and papers

Extracted_Metric__c	Child (Master-Detail)	Metric_Name__c, Value__c, Unit__c, Test_Conditions__c, Research_Paper__c	Numerical benchmark results extracted from papers
Research_Finding__c	Child (Master-Detail)	Claim_Summary__c, Verbatim_Quote__c, Page_Number__c, Confidence_Score__c	Verbatim empirical evidence with page audit trail
Research_Conflict__c	Lookup Object	Contradiction_Summary__c, Paper_A__c, Paper_B__c, Severity__c	Flags direct scientific contradictions between papers
Research_Gap__c	Child (Master-Detail)	Question__c, Proposed_Validation_Experiment__c, Status__c	Unsolved research questions & experimental protocols

4. All Platform Features & UI Components

• **Executive Dashboard (ereunaDashboardHome):** Real-time metric cards, full-width research topic table, and a dedicated top-bar operational menu for immediate manual access.

• **Literature Synthesis Engine (ereunaTopicSearch):** Allows entering any research query across CS, Mechanical, Civil, EEE, ECE, or Chemical Engineering, executing a 4-step synthesis pipeline in 2 seconds.

• **Literature Comparison Matrix (literatureCompareMatrix):** Side-by-side benchmark comparison matrix normalizing metrics across papers without opening spreadsheets.

• **Evidence Inspector (findingsExplorer):** Displays verbatim quotes with exact page numbers, confidence percentages, and methodology tags.

• **Research Gap Kanban Board (gapKanbanBoard):** Drag-and-drop tracker shepherding unsolved problems across 4 lifecycle stages: Identified -> In Review -> Experiment Drafted -> Addressed.

• **User Manual & Governance Tab (governanceManualView):** Full-page documentation tab directly on the Salesforce navigation bar with downloadable PDF resources.

5. Step-by-Step Live Demo Script for Presentations

Follow this exact 3-minute sequence when presenting to examiners or judges:

1. **App Launcher:** Click the 9 dots in Salesforce, select 'Research Radar'. Show the clean, high-contrast monochrome dashboard.
2. **Discipline Filter:** Click 'Chemical', 'Mechanical', or 'Civil' filter pills in the topics table to show multi-branch versatility.
3. **Synthesize:** In the synthesis bar, type *'Serverless cold-start latency'* and click 'Generate Research Map'. Show the progress animation.
4. **Comparison Matrix:** Click 'Open Workspace ->' to show benchmark numbers from multiple papers side-by-side.
5. **Verified Evidence:** Click any finding to prove that verbatim quotes and DOI page numbers prevent AI hallucinations.
6. **Gap Board & Manual:** Show the Kanban board turning gaps into experiments, and point out the top-bar 'User Manual & Legal' tab.

6. Common Viva Questions & Winning Answers

Q1: Why not just use Google Scholar?
Answer: Google Scholar only returns links to PDFs. You still have to spend weeks reading them and making spreadsheets. Research Radar extracts the numbers, compares them side-by-side, flags contradictions, and identifies unsolved research gaps automatically.

Q2: Why not just ask ChatGPT?
Answer: LLMs hallucinate fake citations, wrong numbers, and non-existent authors. Research Radar uses a deterministic relational database where every finding links directly to a publisher DOI with exact page numbers.

Q3: What prevents bad data from corrupting the database?
Answer: Native Salesforce Validation Rules on the database layer reject invalid DOIs (must start with '10.') and out-of-range publication years, even if incoming APIs send bad inputs.

Q4: What happens when a paper is deleted?
Answer: Master-Detail relationships automatically cascade-delete linked metrics and findings, maintaining 100% referential integrity

with zero orphaned data.